

Homework 6 Key

Solutions by:
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$$1. \det(S - \lambda I) = (1 - \lambda)[(-1 - \lambda)(- \lambda) - 4] + 2[0 - 2(-1 - \lambda)]$$

$$= \lambda(9 - \lambda^2)$$

$$\lambda = -3, 0, 3$$

$$\underline{\lambda_1 = -3}: \begin{bmatrix} 4 & 0 & 2 \\ 0 & 2 & -2 \\ 2 & -2 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \vec{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$$

$$\underline{\lambda_2 = 0}: \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix}$$

$$\underline{\lambda_3 = 3}: \begin{bmatrix} -2 & 0 & 2 \\ 0 & -4 & -2 \\ 2 & -2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \quad \vec{v}_3 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$$

$$S = \begin{bmatrix} -\frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} -3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ -\frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

$$2. A^T A = \begin{bmatrix} 2 & 1 & 0 & -1 \\ 1 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 1 \\ 1 & 3 \end{bmatrix} \quad \begin{bmatrix} 6 & 1 & | & 8 \\ 1 & 3 & | & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & | & \frac{24}{17} \\ 0 & 1 & | & -\frac{8}{17} \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} \frac{24}{17} \\ -\frac{8}{17} \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} 2 & 1 & 0 & -1 \\ 1 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \end{bmatrix}$$

$$3. a. A^T A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & | & 2 \\ 1 & 2 & | & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & | & -1 \\ 0 & 1 & | & 3 \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$$

$$b. A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 & | & 8 \\ 2 & 3 & | & 14 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & | & -2 \\ 0 & 1 & | & 6 \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 14 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}$$

4. a. $\lambda = 0, 0$

$$A^T A = \begin{bmatrix} 0 & 0 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 16 \end{bmatrix}$$

$$A A^T = \begin{bmatrix} 0 & 4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 16 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\sigma = 0, 4$$

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$$\underline{\lambda = 0}: \begin{bmatrix} 0 & 0 \\ 0 & 16 \end{bmatrix} \quad \vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\underline{\lambda = 0}: \begin{bmatrix} 16 & 0 \\ 0 & 0 \end{bmatrix} \quad \vec{u}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\underline{\lambda = 16}: \begin{bmatrix} -16 & 0 \\ 0 & 0 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\underline{\lambda = 16}: \begin{bmatrix} 0 & 0 \\ 0 & -16 \end{bmatrix} \quad \vec{u}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 4 \\ 0 & 0 \end{bmatrix}$$

$$b. \det(A - \lambda I) = \lambda^2 - 4 = 0 \Rightarrow \lambda = \pm 2$$

$$A^T A = \begin{bmatrix} 0 & 1 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 16 \end{bmatrix} \quad A A^T = \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 16 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\sigma = 1, 4$$

$$\lambda = 1: \begin{bmatrix} 0 & 0 \\ 0 & 15 \end{bmatrix} \quad \vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda = 16: \begin{bmatrix} -15 & 0 \\ 0 & 0 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix}$$

$$\sigma = 1, 4$$

$$\lambda = 1: \begin{bmatrix} 15 & 0 \\ 0 & 0 \end{bmatrix} \quad \vec{u}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\lambda = 16: \begin{bmatrix} 0 & 0 \\ 0 & -15 \end{bmatrix} \quad \vec{u}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$5. A^T A = \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} \quad \lambda_1 = 2: \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \quad \vec{v}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\begin{aligned} \det(A^T A - \lambda I) &= (5 - \lambda)^2 - 9 \\ &= \lambda^2 - 10\lambda + 16 \\ &= (\lambda - 2)(\lambda - 8) \\ \lambda &= 2, 8 \end{aligned}$$

$$\lambda_2 = 8: \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\vec{u}_1 = \frac{A \vec{v}_1}{\sigma_1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \\ 0 & 2\sqrt{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\vec{u}_2 = \frac{A \vec{v}_2}{\sigma_2} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$